

Finanzas en R - Notebook Dates

Notebook Dates

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Fechas en R

Current date

Por lo general, siempre debemos partir por obtener la fecha y hora actuales:

```
Sys.timezone()
```

```
[1] "America/Santiago"
```

```
Sys.Date()
```

```
[1] "2026-09-25"
```

```
Sys.time()
```

```
[1] "2026-09-25 19:37:21 -03"
```

Lubridate

La librería más utilizada para lidiar con fechas en R, corresponde a lubridate [enlace](#), que es parte de tidyverse.

```
library(tidyverse)
```

```
now()
```

```
[1] "2026-09-25 19:37:22 -03"
```

String to date

Utilizando los formatos de R por defecto:

```
x <- c("2015-07-01", "2015-08-01", "2015-09-01")
```

```
as.Date(x)
```

```
[1] "2015-07-01" "2015-08-01" "2015-09-01"
```

Esto solo funciona debido a que las fechas se encuentran en el formato por defecto. En caso distinto, debemos incorporar el formato de entrada:

```
y <- c("07/01/2015", "07/01/2015", "07/01/2015")
```

```
as.Date(y, format = "%m/%d/%Y")
```

```
[1] "2015-07-01" "2015-07-01" "2015-07-01"
```

Utilizando lubridate

Repliquemos lo anterior:

```
library(lubridate)
```

```
ymd(x)
```

```
[1] "2015-07-01" "2015-08-01" "2015-09-01"
```

```
mdy(y)
```

```
[1] "2015-07-01" "2015-07-01" "2015-07-01"
```

Como ventaja, lubridate debería reconocer de manera automática los separadores normalmente usado en formatos de fecha.

```
z <- c("07012015", "07012015", "07012015")
```

```
mdy(z)
```

```
[1] "2015-07-01" "2015-07-01" "2015-07-01"
```

Crear en base a uniones

```
yr <- c("2012", "2013", "2014", "2015")
```

```
mo <- c("1", "5", "7", "2")
```

```
day <- c("02", "22", "15", "28")
```

```
ISOdate(year = yr, month = mo, day = day)
```

```
[1] "2012-01-02 12:00:00 GMT" "2013-05-22 12:00:00 GMT"
```

```
[3] "2014-07-15 12:00:00 GMT" "2015-02-28 12:00:00 GMT"
```

```
as.Date(ISOdate(year = yr, month = mo, day = day))
```

```
[1] "2012-01-02" "2013-05-22" "2014-07-15" "2015-02-28"
```

¿Cómo haríamos esto con lubridate?

```
ymd(paste0(yr, mo, day))
```

Warning: All formats failed to parse. No formats found.

```
[1] NA NA NA NA
```

¿Cuál es el error?

```
ymd(paste0(yr, '0', mo, day))
```

```
[1] "2012-01-02" "2013-05-22" "2014-07-15" "2015-02-28"
```

¿Esto es una solución definitiva?

Modifiquemos los vectores de fecha:

```
yr_01 <- c("2012", "2013", "2014", "2015", "2023")
mo_01 <- c("1", "5", "7", "2", "12")
day_01 <- c("02", "22", "15", "28", "15")
```

Repetimos el código anterior:

```
ymd(paste0(yr_01, '0', mo_01, day_01))
```

Warning: 1 failed to parse.

```
[1] "2012-01-02" "2013-05-22" "2014-07-15" "2015-02-28" NA
```

Esto es un caso particular, que por lo general lleva a errores. Añademos una función de leading zeros:

```
library(stringr)
ymd(paste0(yr_01, str_pad(mo_01, 2, pad = "0"), day_01))
```

```
[1] "2012-01-02" "2013-05-22" "2014-07-15" "2015-02-28" "2023-12-15"
```

Cabe mencionar que `str_pad` permite añadir ya sean 0 a la izquierda, a la derecha, o ambos lados.

¿Dudas?

Obtener partes de una fecha:

```
x <- c("2015-07-01", "2015-08-01", "2015-09-01")
```

```
year(x)
```

```
[1] 2015 2015 2015
```

```
month(x)
```

```
[1] 7 8 9
```

```
month(x, label = TRUE)
```

```
[1] jul ago sept
```

12 Levels: ene < feb < mar < abr < may < jun < jul < ago < sept < ... < dic

```
month(x, label = TRUE, abbr = FALSE)
```

```
[1] julio      agosto      septiembre
```

12 Levels: enero < febrero < marzo < abril < mayo < junio < ... < diciembre

```
wday(x, label = TRUE, abbr = FALSE)
```

```
[1] miércoles sábado    martes
```

7 Levels: domingo < lunes < martes < miércoles < jueves < ... < sábado

Modificar fechas

```
x <- ymd(x)
```

```
x
```

```
[1] "2015-07-01" "2015-08-01" "2015-09-01"
```

```
mday(x)
```

```
[1] 1 1 1
```

```
mday(x) <- c(3, 10, 22)
```

```
x
```

```
[1] "2015-07-03" "2015-08-10" "2015-09-22"
```

```
update(x, year = c(2013, 2014, 2015), month = 9)
```

```
[1] "2013-09-03" "2014-09-10" "2015-09-22"
```

```
x + years(1) - days(c(2, 9, 21))
```

```
[1] "2016-07-01" "2016-08-01" "2016-09-01"
```

Crear secuencias de fechas

Utilizando la librería básica de fechas:

```
seq(as.Date("2010-1-1"), as.Date("2015-1-1"), by = "years")
```

```
[1] "2010-01-01" "2011-01-01" "2012-01-01" "2013-01-01" "2014-01-01"
```

```
[6] "2015-01-01"
```

```
seq(as.Date("2015/1/1"), as.Date("2015/12/30"), by = "quarter")
```

```
[1] "2015-01-01" "2015-04-01" "2015-07-01" "2015-10-01"
```

```
seq(as.Date('2015-09-15'), as.Date('2015-09-30'), by = "2 days")
```

```
[1] "2015-09-15" "2015-09-17" "2015-09-19" "2015-09-21" "2015-09-23"
```

```
[6] "2015-09-25" "2015-09-27" "2015-09-29"
```

Utilizando lubridate:

```
seq(ymd("2010-1-1"), ymd("2015-1-1"), by = "years")
```

```
[1] "2010-01-01" "2011-01-01" "2012-01-01" "2013-01-01" "2014-01-01"
```

```
[6] "2015-01-01"
```

```
seq(ymd("2015/1/1"), ymd("2015/12/30"), by = "quarter")
```

```
[1] "2015-01-01" "2015-04-01" "2015-07-01" "2015-10-01"
```

```
seq(ymd('2015-09-15'), ymd('2015-09-30'), by = "2 days")
```

```
[1] "2015-09-15" "2015-09-17" "2015-09-19" "2015-09-21" "2015-09-23"
```

```
[6] "2015-09-25" "2015-09-27" "2015-09-29"
```

En el caso de añadir tiempo, los formatos por defecto de R deben considerar que sean tipo POSIXct y no Date (producidos por `as.Date`)

```
seq(as.POSIXct("2015-1-1 0:00"), as.POSIXct("2015-1-1 12:00"), by = "hour")
```

```
[1] "2015-01-01 00:00:00 -03" "2015-01-01 01:00:00 -03"
```

```
[3] "2015-01-01 02:00:00 -03" "2015-01-01 03:00:00 -03"
```

```
[5] "2015-01-01 04:00:00 -03" "2015-01-01 05:00:00 -03"
```

```
[7] "2015-01-01 06:00:00 -03" "2015-01-01 07:00:00 -03"
```

```
[9] "2015-01-01 08:00:00 -03" "2015-01-01 09:00:00 -03"
```

```
[11] "2015-01-01 10:00:00 -03" "2015-01-01 11:00:00 -03"
```

```
[13] "2015-01-01 12:00:00 -03"
```

Con `lubridate`:

```
seq(ymd_hm("2015-1-1 0:00"), ymd_hm("2015-1-1 12:00"), by = "hour")
```

```
[1] "2015-01-01 00:00:00 UTC" "2015-01-01 01:00:00 UTC"
```

```
[3] "2015-01-01 02:00:00 UTC" "2015-01-01 03:00:00 UTC"
```

```
[5] "2015-01-01 04:00:00 UTC" "2015-01-01 05:00:00 UTC"
```

```
[7] "2015-01-01 06:00:00 UTC" "2015-01-01 07:00:00 UTC"
```

```
[9] "2015-01-01 08:00:00 UTC" "2015-01-01 09:00:00 UTC"
```

```
[11] "2015-01-01 10:00:00 UTC" "2015-01-01 11:00:00 UTC"
```

```
[13] "2015-01-01 12:00:00 UTC"
```